



FOUNDATION UNIVERSITY AND SCHOOL OF SCIENCE AND TECHNOLOGY

Department of Electrical and Computer Engineering

Data Structure and Algorithm Project

Project Title:

Online Order Processing System using Data Structures in C++

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Introduction :

Using Data structure Concept in C++ With Single Linked list to store and manage all the orders dynamically . Each order is stored in node form and all details of order are connected through pointer addressing .

The purpose of this is stores and manage the orders for any business Like we use to make it for a Coffee shop as Managing orders are to much time consuming and difficult to solve this problem So now this sytem provides menu based interface for customer

Objectives :

The main objectives are :

- To mange customer orders efficiently
- Implementation of DSA in real life

- To reduce manually handling orders
- To understand the concepts of programming and solving real life complex problem

Features of the System :

- Place New order
- Show all orders
- Show waiting or pending orders
- Remove order when completed
- Exit

Data Structure Used:

- This project uses Singly linked list
- Each node stores data and pointer
- Pointers connects different nodes
- The last node is null

Why Linked List?

- Order can be added dynamically
- Memory used efficiently
- Not Fixed Size

System module :

- First Place order
- Show all orders
- Waiting orders
- Complete orders
- Remove orders
- Exit

Working Principle (Customer Side):

- Select Product from provided menu
- System places order and gives order no
- Nodes are connected until user exists

Algorithm :

- Input customer details
- Created nodes using order details if list is empty
- Otherwise Traverse new node

Advantages :

- Easy to use
- No need to remember order details
- Fast insert and delete Order

Limitations :

- No data base
- Only console based
- No much security

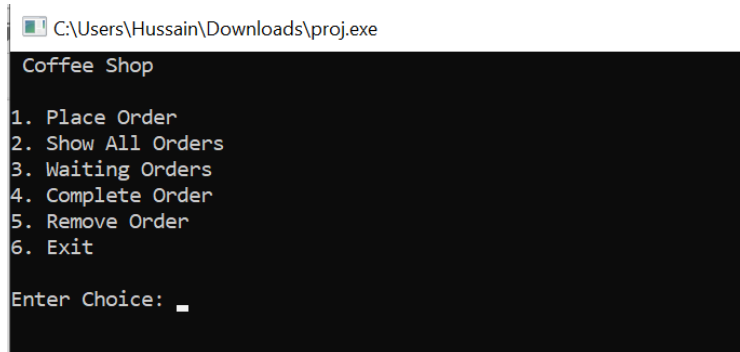
Future Improvement :

- Database integration
- File handling for History and stats
- GUI
- Billing system

Conclusion :

This coffee shop Order management shows how easily we can Provide any business Such a type of C++ Singly linked List based system that works efficiently and reduce the time of taking orders manually And it also hepls to understand the concept of DSA like Linked list pointers, traversal, nodes ,insertion, and deletion.

Console Interface :

A screenshot of a Windows console window. The title bar shows the file path "C:\Users\Hussain\Downloads\proj.exe". The console content displays a menu for a "Coffee Shop" with six numbered options: "1. Place Order", "2. Show All Orders", "3. Waiting Orders", "4. Complete Order", "5. Remove Order", and "6. Exit". Below the menu, the prompt "Enter Choice: " is followed by a cursor and a space character.

```
C:\Users\Hussain\Downloads\proj.exe
Coffee Shop
1. Place Order
2. Show All Orders
3. Waiting Orders
4. Complete Order
5. Remove Order
6. Exit
Enter Choice:  
```

Source code Repository link :

https://github.com/alihussain0202/DSA_PROJECT/tree/main